

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5102

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct: (192521302)

I A. Midhuna (192521302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

A. Midhuna

Signature of the student:

85
100

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
 - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
 - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
 - a. Deduce the most likely original message.
 - b. Justify your answer by analysing repeating patterns, letter structures or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
 - a. Demonstrate how the encryption process transforms the message step-by-step.
 - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
 - a. Analyze how investigators could detect the presence of hidden data within the image.
 - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".
 - a. Determine the most probable original message by reversing the encryption process.
 - b. Critically evaluate why such a method is vulnerable in modern communication systems.
 - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

A. Mldhuna

Application - Based questions

192521302

5/10/26

Scenario: Event Communication Security: A student organization preparing to share confidential event details with its core members using a simple classical encryption method. During testing a message "MEET AT NOON" transformed into an encoded version using fixed shift pattern.

- Reconstruct how encrypted message might appear shift 4 positions applied.
- Explain how authorized member aware encoding rule could retrieve original message.

Introduction:

The given scenario uses caesar cipher classical substitution encryption technique. In this method each letter message shifted fixed number of position alphabet. The fixed shift 4 positions.

a) Encrypted message

MEET AT NOON

Shift each letter 4 positions forward in the alphabet:

plaintext

ciphertext

M

Q

E

I

E

I

T	X
A	E
T	X
N	R
O	S
O	S
N	R

Plaintext: MEET AT NOON

Encrypted Message: ~~011X~~ EX RSSR

Spaces between words are not changed.

b) Decryption of message

An authorized member who knows encryption rule & shift value decrypt the message

For eg: $01 \rightarrow m$

$1 \rightarrow E$

$1 \rightarrow E$

$X \rightarrow T$

$E \rightarrow A$

$X \rightarrow T$

$R \rightarrow N$

$S \rightarrow O$

$S \rightarrow O$

$R \rightarrow N$

Ciphertext)

~~011X~~ EX RSSR

After shifting:

MEET AT NOON

conclusion:

The caesar cipher provide simple protect event message
 fixed no of positions. "MEET AT NOON" becomes "011X EX RSSR"
 in authorized member shift value shift letter 4 position backward.

Scenario: Intercepted suspicious message: A cybersecurity intern intercepts coded message: "GISRH RH Z HVXIVGI" during network monitoring. The structure of text suggest simple substitution mechanism rather complex encryption.

a) deduce most likely original message.

b) Justify your answer analysis repeating patterns, letter structures known cipher characteristics.

Introduction:

The coded message is "GISRH RH Z HVXIVGI"

The structure message suggest use simple substitution cipher. The most like cipher Atbash cipher alphabet reversed.

Substitution is

$A \leftrightarrow Z, B \leftrightarrow Y, C \leftrightarrow X, D \leftrightarrow W, \dots, M \leftrightarrow N$

a) deduce original message

coded Letter	original Letter
G	T
S	H
R	I
H	S
R	I
H	S
Z	A
H	S

I	R
V	E
G	T

ciphertext: GSRH RH Z HVXIVG plaintext: THIS IS A SECRET

b) Justification

Repeating word pattern

coded message: GSRH RH Z HVXIVG

word length of decrypted

THIS IS A SECRET

2. Repeated letters:

The coded word GSRH contain four different letters

G → T

GSRH → THIS

S → H

R → I

one letter word:

H → S

ciphertext contains one letter word Z

Z → A

Known Atbash characteristic

Plain : A B C D E F G H I J K L M

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
cipher : Z Y X W V U T S R Q P O N

using rule: GSRH RH Z HVXIVG becomes: THIS IS A SECRET

conclusion:

The 4-2-1-6 word pattern one letter word Z → A

Consistent letter substitutions reversed alphabet property

Atbash cipher all support conclusion.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

20

20

20

15

10

85